CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of BSc (BT/MI/BC) Examination April-May 2018

Course code & Name: BS601 Bioinformatics

Date: 03/05/2018 Day: Thursday Time: 1:30 pm To 1:50 pm Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions. **10**

1. Which of following is not protein composite database?

(i) PROSITE (ii) PIR

(iii) Pfam (iv) PRINTS

2. _________ is the process of identifying and assigning the locations of genes in coding regions of a genome.

(i) genome annotation (ii) genome analysis

(iii) Gene expression (iv) Gene tagging

3. Which of the following is not correct about BLAST?

(i) The BLAST web server has been designed in such a way as to simplify the task of program selection.

(ii) The programs are organized based on the type of query sequences

(iii) BLAST is faster than FASTA

(iv) BLAST is not based on heuristic searching methods

4. In Multiple sequence alignment * (asterisk) indicates positions which is_____.

(i) fully conserved residue (ii) partial similar residue

(iii) dissimilar residue (iv) None of these

5. Which secondary structure is formed when residues have negative value of ϕ (phi) angle and positive value of ψ (psi) angle?

(i) Antiparallel β sheet (ii) Parallel β sheet

(iii) Collagen triple helix (iv) All of these

6. In CATH, TIM barrel, sandwich and roll represents _________of protein classification.

(i) Class (ii) Architecture

(iii) Topology (iv) Homologous superfamily

7. Which of the following is/are protein structural alignment tool?

(i) Cn3D (ii) RASMOL

(iii) DALI (iv) (i) and (iii) both

8. BLOSUM matrices are used for ____________.

(i) phylogenetic analysis (ii) multiple sequence alignment

(iii) pairwise sequence alignment (iv) none of these

9. In the phylogenetic tree, __________represents a single present-day taxon (current spices).

(i) root (ii) node

(iii) leaf (iv) Internode

10. Genes are similar due to convergent evolution, termed as______.

(i) homologous (ii) paralogous

(iii) orthologous (iv) analogous

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VI Semester of BSc (BT/MI/BC) Examination April-May 2018

Course code & Name: BS601 Bioinformatics

Date: 03/05/2018 Day: Thursday Time: 1:51 pm To 3:30 pm Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed (any five) 10

1. Explain in brief about different sequence format and enlist the tools used for sequence format conversion tools.
2. Write a note on "Protein Data Bank".
3. What is data mining? Draw the flow of data mining process.
4. Write the differences between Pairwise and multiple sequence alignment.
5. Write a note on "Consensus tree".
6. Define: Accession number and query coverage

SECTION - II

Q-III Answer the following questions as directed (any five) 15

1. What is MSA? Explain different steps of MSA procedure.
2. What is Data database? What is need of bioinformatics databases? Explain in detail about primary protein database.
3. Explain Fold, Domain and Superfamily with appropriate example for structural classification of protein.
4. Explain in brief about BLAST N, BLAST P and BLAST X.
5. What is bioinformatics? Explain its application in pharmacogenomics, and gene prediction.

6. Analyse the following scoring of UPGMA cluster and draw the phylogenetic tree. Here, a,b,c,d and e are 16s rRNA gene sequences from different organisms.

	a	b	c	d	e
a	0	17	21	31	23
b	17	0	30	34	21
c	21	30	0	28	39
d	31	34	28	0	43
e	23	21	39	43	0

1 st clustering

	(a,b)	c	d	e
(a,b)	0	25.5	32.5	22
c	25.5	0	28	39
d	32.5	28	0	43
e	22	39	43	0

2nd clusturing

	((a,b),e)	c	d
((a,b),e)	0	30	36
c	30	0	28
d	36	28	0

3rd clustering

7. Analyze the following dynamic scoring and write the two aligned sequence. (indicate indels and gaps in aligned sequence)

		C	G	G	A	T	C	A	T
	0	-3	-6	-9	-12	-15	-18	-21	-24
C	-3	8	5	2	-1	-4	-7	-10	-13
T	-6	5	3	0	-3	7	4	1	-2
T	-9	2	0	-2	-5	5	-1	-1	9
A	-12	-1	-3	-5	6	3	0	7	6
A	-15	-4	-6	-8	3	1	-2	8	5
C	-18	-7	-9	-11	0	-2	9	6	3
T	-21	-10	-12	-14	-3	8	6	4	14